Tanya Robinson

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Summary

Senior Machine Learning Engineer with over 11 years of experience designing, training, and deploying scalable AI systems and data pipelines. Skilled in developing end-to-end ML solutions—from data ingestion and feature engineering to production-grade model deployment. Proven success delivering measurable business impact in e-commerce, fintech, and intelligent transportation, combining strong software engineering expertise with deep data science knowledge and applied research experience.

Skills

- **Machine Learning:** LSTM, BERT, Transformers, Contrastive Learning, Flag Embedding, CRF-NER, Time-Series Forecasting, Recommendation Systems, Semantic Search, Query Understanding, Image-to-Product Retrieval, Object Detection, Edge AI (Raspberry Pi, ARM Cortex)
- **Programming & Frameworks:** Python, Golang, TensorFlow, TensorFlow Lite, PyTorch, ONNX, FastAPI, Spark, Airflow, MLFlow, Katib, Feast
- **Data Engineering & Infrastructure**: ETL, Data Pipelines, Vector Databases, Redis, Kafka Streams, DataLake, Protobuf, GitLab CI/CD
- **Software & DevOps:** Microservices, API Development, Model Benchmarking, Real-Time Analytics, Prometheus, Grafana
- **Domain Expertise:** E-commerce Search & Recommendation, Intelligent Transportation (ETA, GPS Modeling), FinTech, Healthcare / Patient Monitoring, Customer Support Chatbots
- **Other:** Mentorship, Team Leadership, Experimentation & A/B Testing, Model Versioning & Deployment, KPI-driven Model Evaluation (MAPE, R^2, F1, Conversion Rate)

Education

University of Texas at Austin, Austin, TX — May 2014

Bachelor of Computer Science,

Professional Experience

Senior Machine Learning Engineer — July 2023–Present

Instacart, San Francisco, California

- Applied **Contrastive Learning** and **Flag Embedding** methods to improve fine-tuning efficiency of semantic search models, reducing model training time by **30%**.
- Developed and trained a **Query Understanding (QU) model** using user query logs and clickstream data to filter low-confidence queries, decreasing **zero-click searches by 4%**.
- Enhanced **search relevance** through multi-metric fine-tuning and **category-consistent retrieval** using Category Loss and GMV-weighted optimization, increasing **conversion rate (CR) by 4%**.
- Designed a **Claude-powered query variation generation system** to strengthen model generalization and long-tail recall, reducing zero-click searches by **5%**.
- Led the **Exact Match** initiative to improve product code recognition (e.g., “PG32UCD Laptop”) by modifying tokenizers, implementing custom masking, and creating a **Redis-based code index**, reducing seller complaints by **20%**.
- Researched **search feedback loops** leveraging real-time engagement signals and evaluated **ANCE embeddings** for continuous semantic updates.
- Built a **scalable user segmentation pipeline** using embedding-based clustering and feature engineering, boosting conversion rate by **3%** through refined premium user targeting.
- Benchmarked and optimized **Semantic Search** versus **Elasticsearch** pipelines across high-volume queries to guide intelligent routing, improving **add-to-cart rate by 3%**.
- Developed an **LLM-powered chatbot** for product discovery and recommendation; integrated semantic retrieval and contextual ranking pipelines, increasing user engagement time by **5%**.
- Built an **image-to-product retrieval system** combining **CLIP-based dual-encoder models** with textual product embeddings, achieving **92% Top-10 accuracy** on similarity search benchmarks.
- Integrated an **object detection pipeline** for visual search to localize products in uploaded images, improving matching precision by **3%** and reducing false positives by **4%** using region-based embeddings.

Technologies: Python, PyTorch, TensorFlow, LLMs, Retrieval Systems, Vector Search, ElasticSearch, Redis, Data Engineering, Experiment Design, A/B Testing, Statistical Modeling

Senior Data Engineer Dec 2020–June 2023
Iteris, Inc. Austin, TX

- Built a **recommendation system** to estimate street speeds for low-data areas, expanding coverage from **1M to 3M streets**.
- Developed **ETA prediction models** for 8+ cities in California and Washington, improving **R^2 performance by 20%**.
- Collaborated with cloud engineers to create a **full ML pipeline** for training, versioning, and deployment using **Airflow, Spark, MLFlow, Katib, Feast, TensorFlow Serving, FastAPI, Kafka, and GitLab CI/CD**, reducing model training and deployment time by **70%**.
- Developed a **Golang microservice** for real-time model benchmarking, logging metrics with **Prometheus**, and creating dashboards in **Grafana**, speeding up QA processes by **80%**.
- Partnered with product teams to cluster 40+ cities into 10 optimized regions, reducing the total number of models required.
- Implemented **HMM algorithms** to match GPS probes and calculate driver speeds, enabling more accurate traffic modeling.
- Trained and deployed predictive models for street speed in future time buckets, reducing **ETA MAPE by 5%**.
- Led a project to suggest **optimal pickup locations** for drivers and passengers, decreasing offer-to-accept time by **5%**.
- Integrated **vector databases** for similarity search in shop, improving item recommendation conversion rate by **5%**.
- Fine-tuned a pre-trained **OCR model (EasyOCR)** to automate driver ID verification, cutting signup process time from days to hours.
- Used **ONNX** to accelerate transformer model inference by **10%**, separating training and deployment workflows.
- Engineered a **sentiment analysis service** using **SVM**, processing 10,000+ Tweets daily to deliver real-time insights on public sentiment.
- Mentored 5+ new engineers, launched a structured **mentorship program**, and established a **new technical interview pipeline** to improve hiring consistency.

Technologies: Collaborative filtering, Content-based filtering, Time-series forecasting, RNN, SARIMA, Tree-based models, Hierarchical clustering, Unsupervised learning, predictive modeling

Machine Learning Engineer — Oct 2017–Nov 2020
ClearBlade, Austin, TX

- Fine-tuned and deployed **LLM models on GPU** for real-time text translation between support teams and foreign customers.
- Developed a **CRF-based Named Entity Recognition (NER) model** for address search, increasing successful search matches by **15%**.
- Spearheaded the development of a **stacked LSTM model** to predict stock and values, achieving **87% prediction accuracy**.
- Designed and implemented a **type-ahead search system** for stock queries using **prefix matching and a custom trie data structure**, reducing search time by **30%**.
- Built an **AI-driven BERT chatbot** to assist users with stock inquiries, improving engagement and facilitating investment decisions, increasing **user engagement by 20%**.
- Deployed **neural network models for time-series forecasting** on **Raspberry Pi 4** and **decision tree classifiers on ARM Cortex-M52**, enabling edge AI solutions.
- Classified patients into 20+ categories using **active learning**, improving model accuracy for healthcare monitoring applications.
- Applied **Kalman filtering** to enhance GPS positioning by **10%** for precise location tracking.
- Implemented **time-series forecasting models** to predict user location in 30-minute intervals and trigger alerts for deviations.
- Optimized models using **TensorFlow Lite**, reducing memory usage by **50%** for mobile and embedded deployment.
- Built a **high-throughput Kafka pipeline** using **Protobuf** to ingest sensor data into a **DataLake** handling **20k+ messages per second**.
- Designed and deployed an **event detection system** for patient abnormal behavior, achieving **0.95 F1 score**.

Technologies: Python, BERT, Raspberry Pi, LLM, NER, LSTM, Kalman, Cortex, TensorFlow

Software Developer Nov 2014–Aug 2017

Kodexo Labs, Austin, TX

- Developed and maintained **dashboard applications** using **Flask**, HTML, CSS, and JavaScript to support internal tools and dashboards.
- Assisted in **training and fine-tuning LLM models** for multilingual customer support and chatbot applications.
- Built **REST APIs** to serve ML models in production using **Flask** and **FastAPI**, enabling real-time predictions.
- Collaborated on **recommendation and search features**, implementing basic vector-based similarity search for e-commerce items.
- Designed **data pipelines** with Airflow and Kafka to automate data ingestion, processing, and model deployment workflows.
- Wrote clean, maintainable code, performed code reviews, and followed **Agile/Scrum** development practices.
- Contributed to **front-end UI improvements** and dashboards to visualize analytics metrics and ML predictions.

Technologies: Python, Flask, Airflow, Kafka, Agile, HTML
